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E-wallet Analysis Report

Droppping Bill Payment Success Rate

Outline

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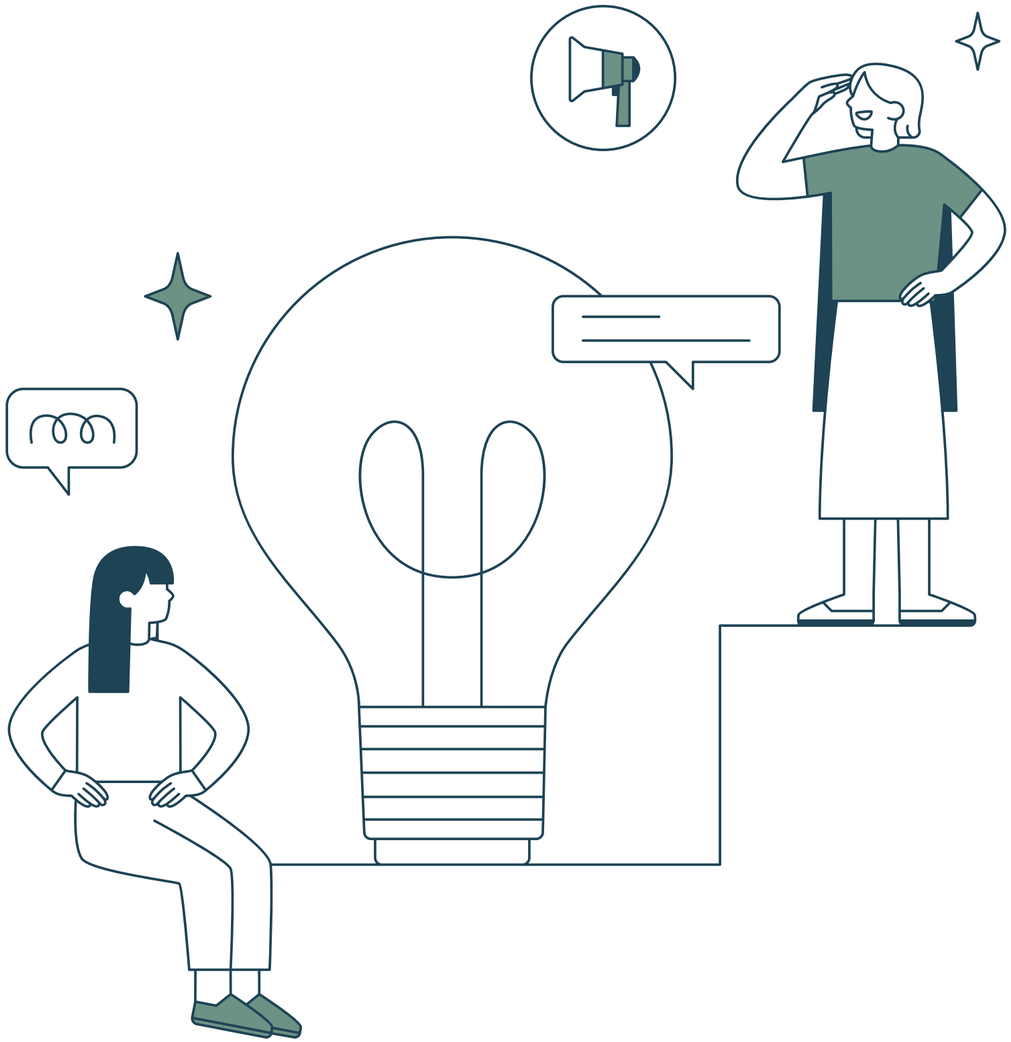


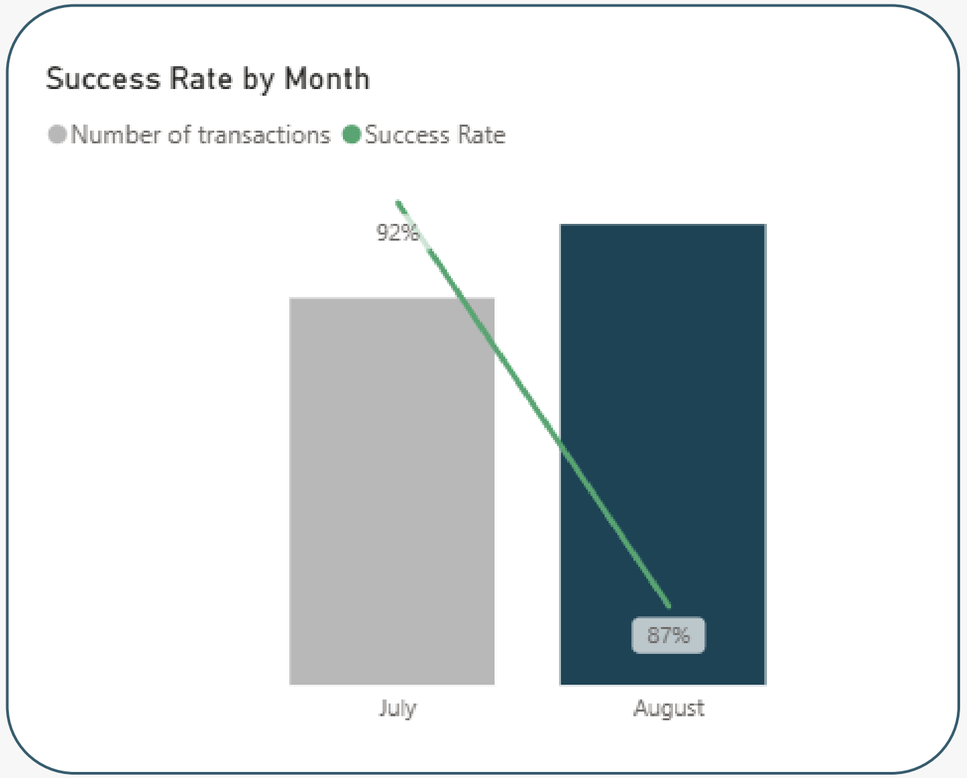
Executive Summary

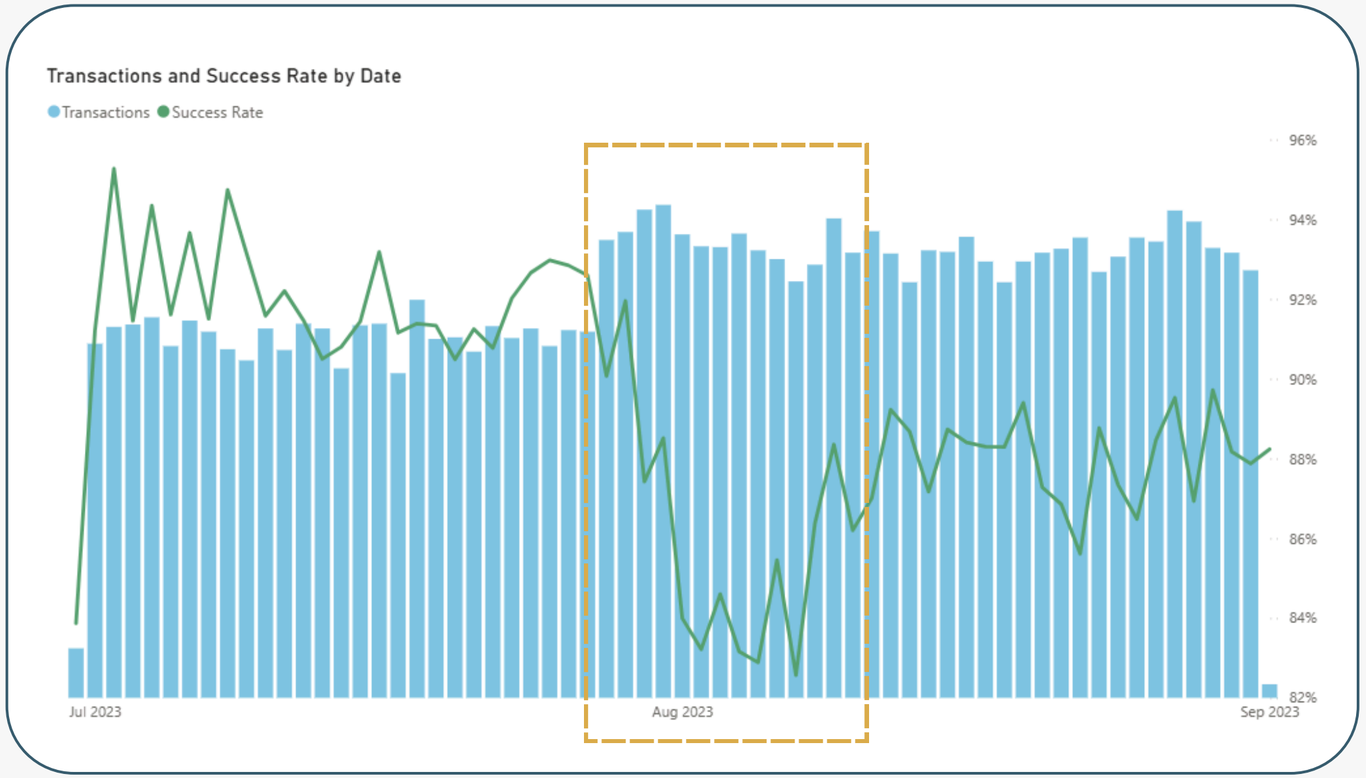
SITUATION **Business Context**	• Transactions increased after the 3.9.6 update on July 28th . • Billing payment transaction accounts for the greatest share (37%).
COMPLICATION **Business problems**	• **Declining Success Rates:** Overall payment success rate dropped by 5% between July and August. • **Authentication Bottlenecks:** High drop-off rates driven by 3DS and OTP timeouts, specifically during the authentication phase. • **Card Performance Issues:** Credit and Debit cards underperformed other methods with a peak 14% error rate. • **Segment Risk:** Dormant users experienced disproportionately higher error rates compared to active users.
QUESTIONS **Business concerns**	• How can we stabilize the backend infrastructure and optimize the authentication flow to reduce the 18% drop-off? • How can we successfully re-engage our dormant and user segments aged 35-44? • How can we improve the customer support system with in-time communication and insightful information?
ANSWERS **Business solutions**	• Reduce authentication failures and backend-related drop-offs. • Payment flow optimization & Root cause validation by performing A/B testing and comparison between versions. • Improving real-time customer support and re-engaging with dormant users through marketing campaigns.

The payment success rate declined despite an increase in total transactions

The payment success rate **declined from 92% to 87%**, although the number of total transactions increased by approximately **20%** between July and August 2023.

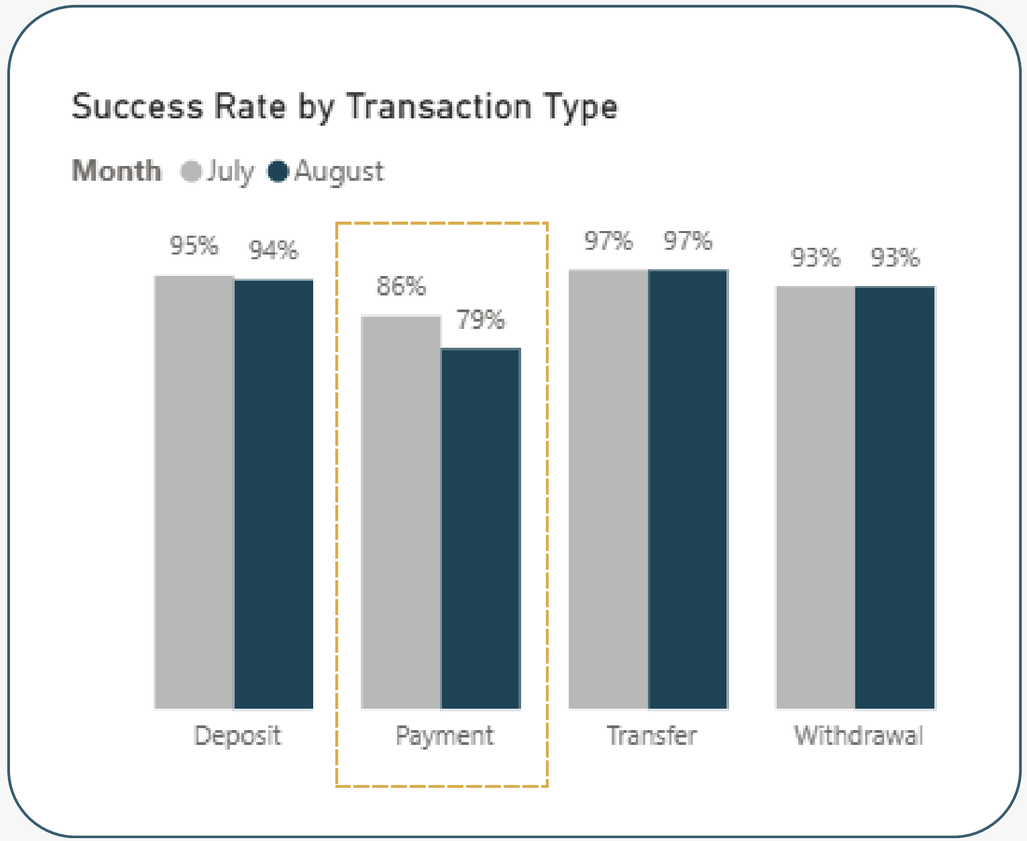
There was a sharp decrease on July 28th, which was followed by a more intense drop until August 8th. However, the **success rate** at the end of the period remained **lower** than in the first half of July.





Billing payment transaction success rate decreased by 7%

Payment transactions saw the highest decline of **7%**, indicating technical or compliance issues related to billers/ providers.



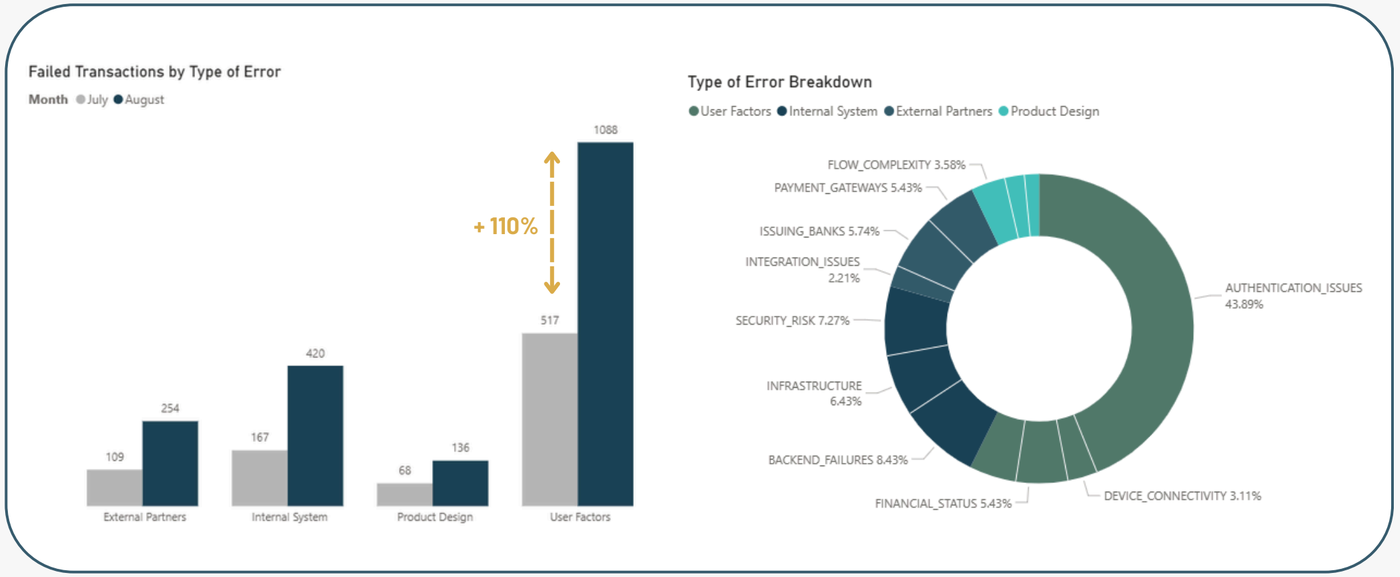
Billing payment transaction accounts for the greatest share (**37%**). However, it decreased significantly by **14%** between July and August, resulting in a substantial drop in the total success rate.

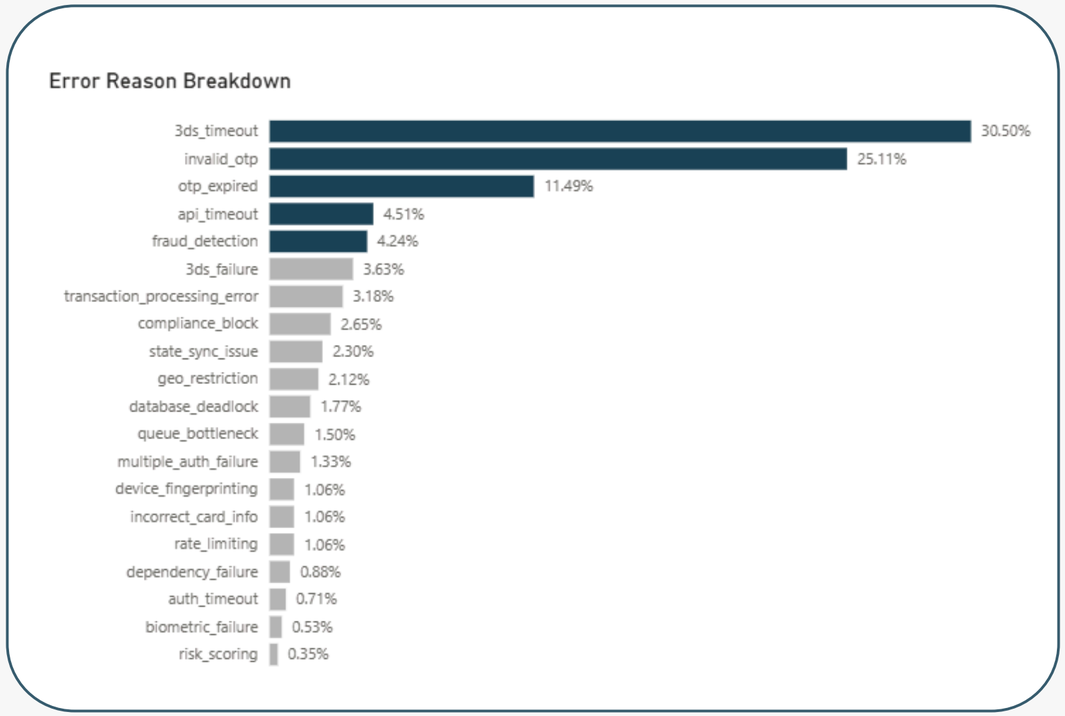
Category	SR_7	SR_8	% Change	% Share
Billing	86%	72%	**-14%**	**37%**
FnB	86%	82%	-4%	24%
Marketplace	87%	85%	-2%	15%
Movies	85%	79%	-6%	11%
Shopping	87%	86%	-1%	7%
Telco	84%	79%	-5%	3%
Transportation	86%	84%	-2%	3%
Traveling	88%	81%	-7%	2%

Authentication issues are the main causes, with timeout errors contributing over 46%

- Errors from user factors increased significantly by (+110%) from July to August.
- **Authentication issues** are the greatest contributors (44%). Meanwhile, **backend failures** and **security risks** are also significant problems with the internal system.

The authentication phase is most affected by time-related failures, with **timeouts across 3DS, OTP delivery, and API responses** accounting for **46%** of issues. This is followed by **invalid OTP errors at 25%.**

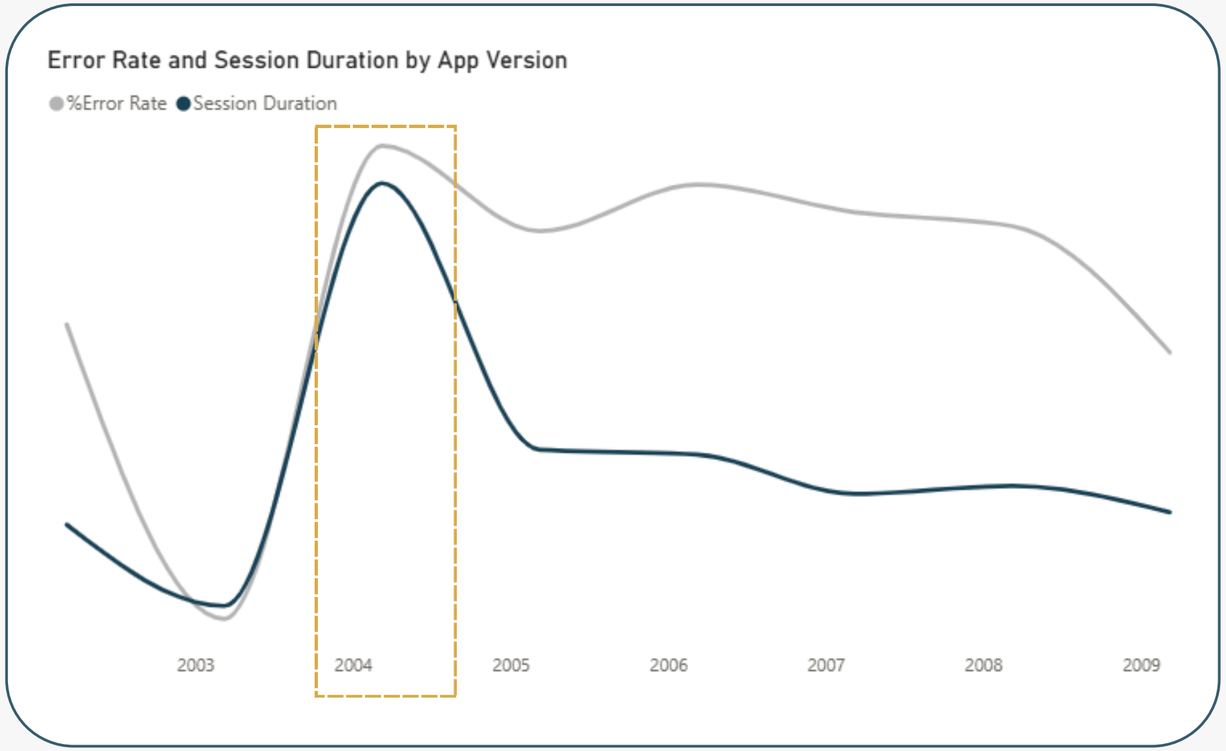


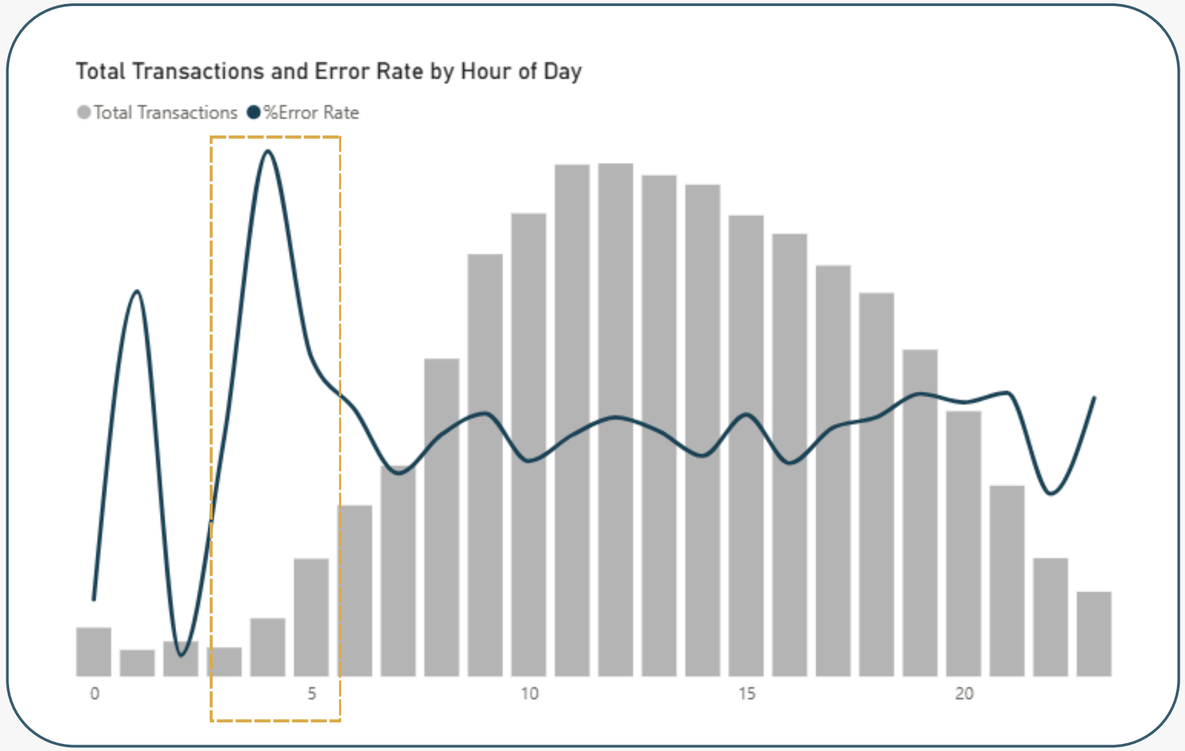


Error rate was highest for the 2004 app version and during early mornning

- Error rate and the amount of time spent on each session are **directly proportional.**
- The highest error recorded was through the **2004 app version.**

The payment error rate **reached its peak at 4 am**, although the total number of transactions was highest during noon.
→ Traffic congestion is not a problem. Rather, the problem might come from system maintenance, infrastructure, or fraud issues without timely support.

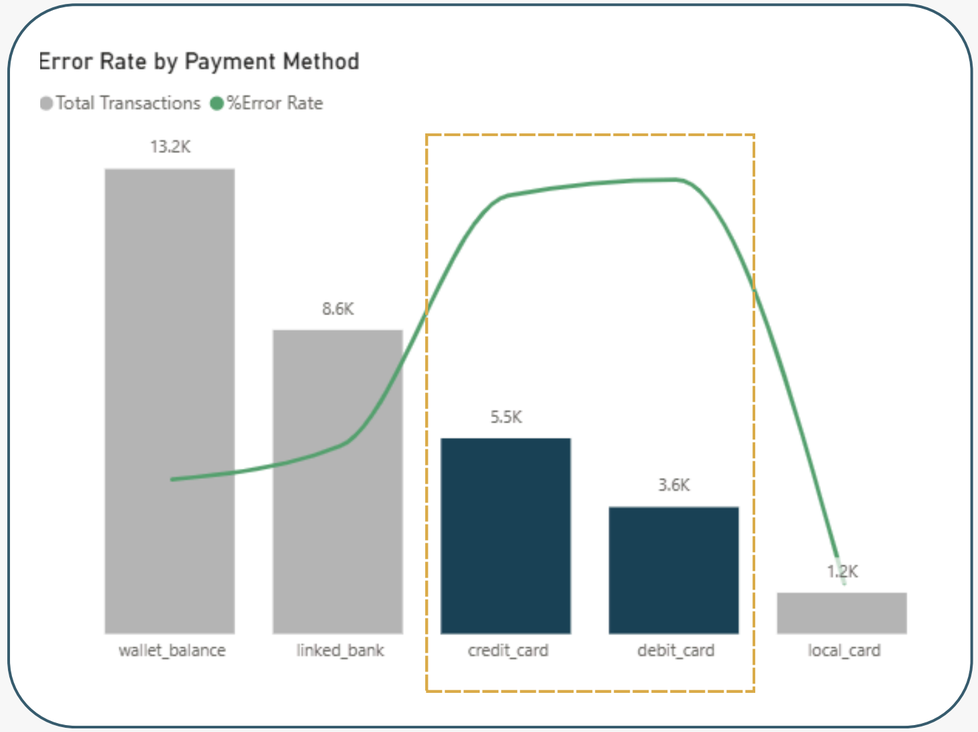


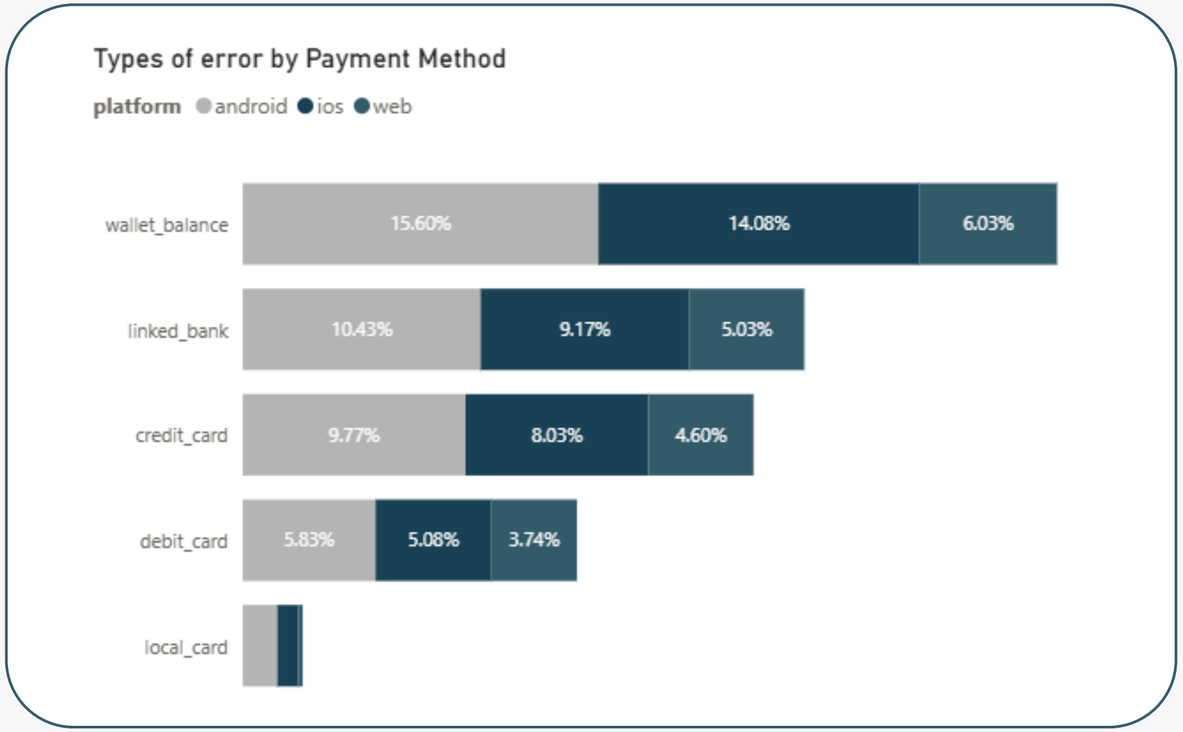


Transactions through Credit Card and Debit Card recorded the highest error rate

- Despite lower transactions than other types, **Credit Card and Debit Card** recorded the highest error rate, at **14%** respectively.
- This signals that there is a problem with payment flow through third parties (banks), consolidating that 3Ds and OTPs authentication needs to be improved.

- Making transactions by phone, through both **iOS and Android** operating systems, led to **80% errors**.
- Meanwhile, transactions through web methods, such as Payment Gateway and QR Code Scanning, are more stable with an error rate under 6%.

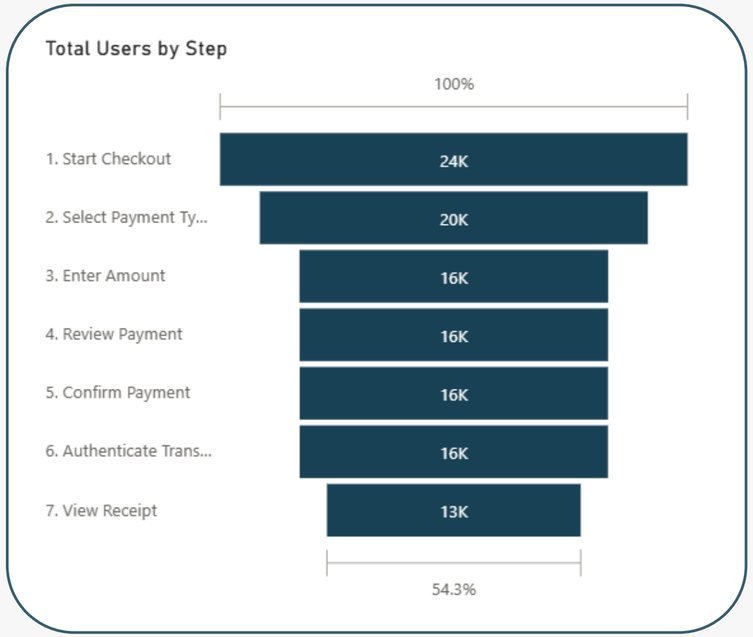




The post-authentication phase recorded the most drop-offs

- The overall conversion rate is **54.3%**, with 13K users successfully reaching the receipt stage.
- The most critical friction point occurs immediately after the app opens, where **17% of users (4K)** exit the process.
- Following the authentication phase, there was a notable **18% drop-off**. Addressing this bottleneck will boost our success rate significantly.

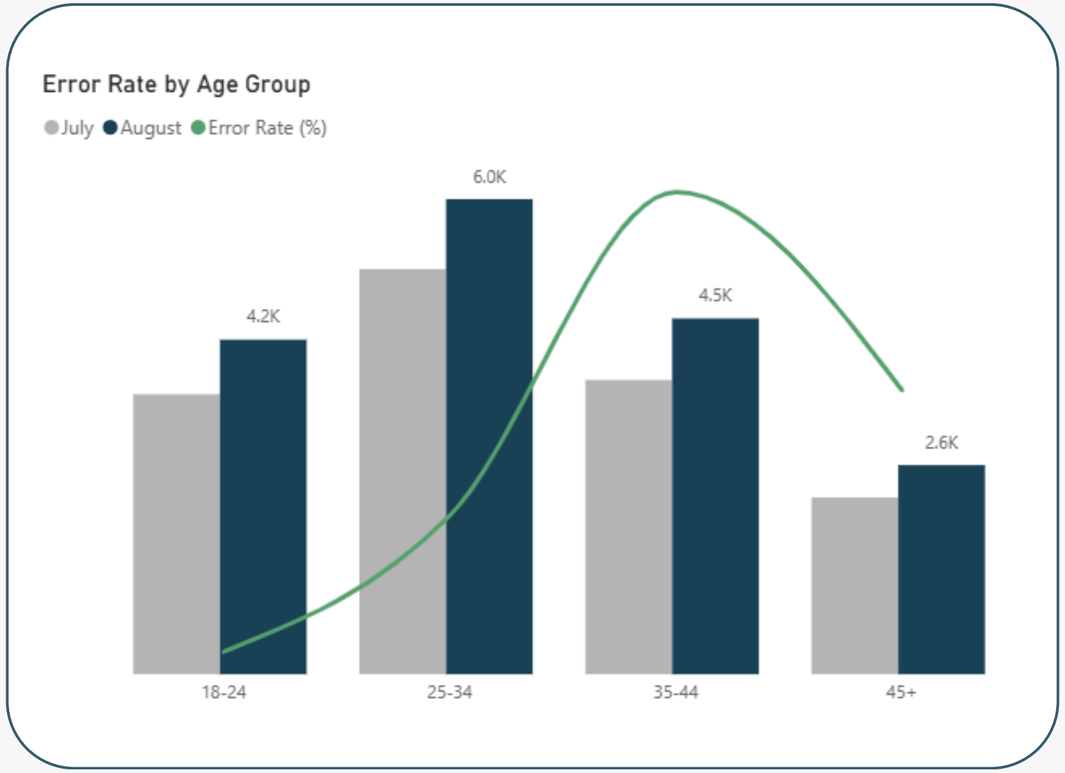
- The total time to convert is **141.53** seconds (~2 minutes 30 seconds).
- Users spend only **~5 seconds on Step 2**, suggesting drop-offs are likely caused by misclicks rather than UX complexity.
- The **Authentication phase has the longest average duration (31.63 seconds)**, though the difference compared to other steps is small (~2 seconds). The latest version (3.9.9) has already addressed this issue by improving payment gateway connectivity and increasing transaction timeout.



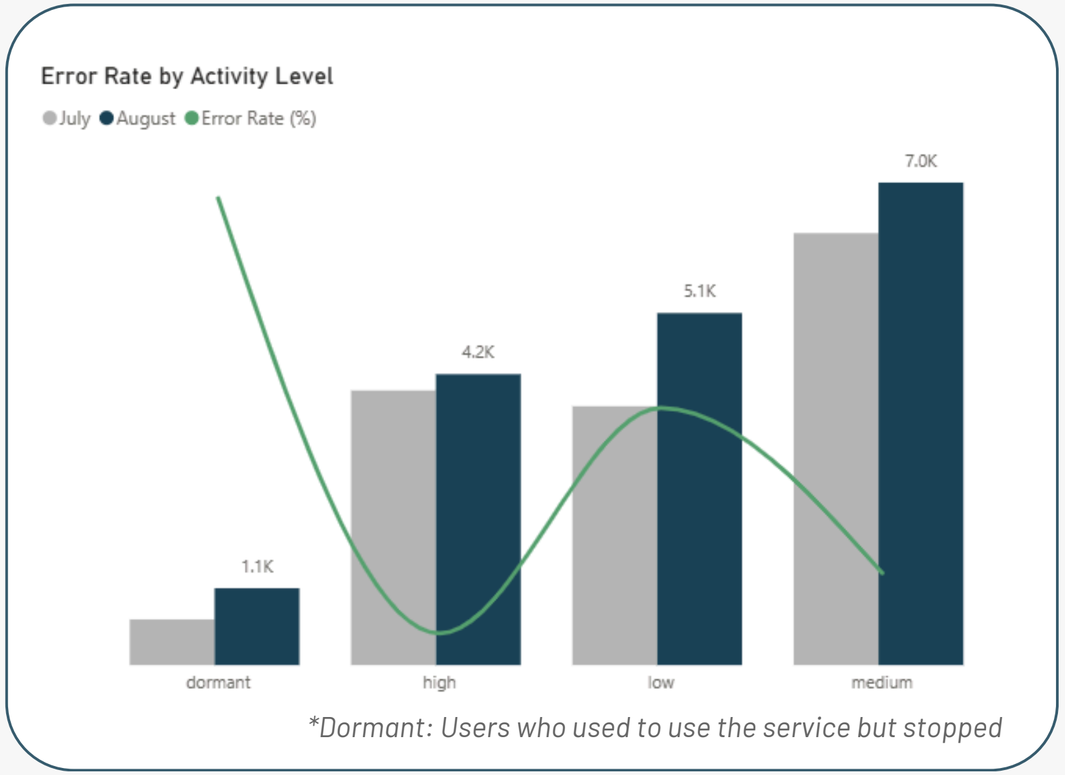
Step	Total Users	%First Step	%Previous Step	Avg time from First Step	Avg Time from Previous Step
1. Start Checkout	23,551	100%		0.00	
2. Select Payment Type	19,551	83%	83%	5.49	5.49
3. Enter Amount	15,551	66%	80%	28.80	23.28
4. Review Payment	15,541	66%	100%	42.69	13.98
5. Confirm Payment	15,532	66%	100%	71.61	28.79
6. Authenticate Transaction	15,531	66%	100%	97.12	31.63
7. View Receipt	12,784	54%	82%	141.53	27.68

The 35-44 age group and Dormant users have the highest error rate

- The **error rate peaked in the 35-44 age group** (11%), while the number of users was lower than in other age brackets.
- The **error rate** tends to **increase gradually** from the youngest group (18-24) and peaks in middle age (35-44) before decreasing in the 45+ group.



- **The most inactive (Dormant) users have the highest error rate (14%)**, indicating that transaction errors affect churn rate.
- High-activity users have the lowest error rate. They may be familiar with the transaction flow and hence, make fewer operational errors.



Recommendations

Reduce authentication failures and backend-related drop-offs

- **Increase OTP session timeout** to reduce authentication expiration.
- Review version 3.9.9 and **fix payment gateway API bugs and backend connectivity failures** if needed.
- Review the fraud detection process to reduce false cases.
- Investigate the 2004 app version, considering suggested updates for users.
- **Conduct card-specific debugging** with billers/providers to identify root causes of Credit/Debit failures during authentication.

Payment flow optimization & Root cause validation

- **Run A/B testing** on the authentication flow (especially post-version 3.9.8 redesign).
- Compare pre- and post-redesign performance metrics to isolate impact.
- **Promote alternative payment methods** (e.g., Wallet Balance) with higher success rates while Card Payment issues are being investigated.

Real-time support and Dormant users re-engagement

- **Provide support for users aged 35+** and increase morning support, suggestively using an AI chatbot.
- In-app real-time error guidance.
- **Retarget users who dropped during authentication** (Dormant users) through email and notifications.
- Considering marketing campaigns to promote new feature updates and educate on the new payment flow.

Thank you for listening